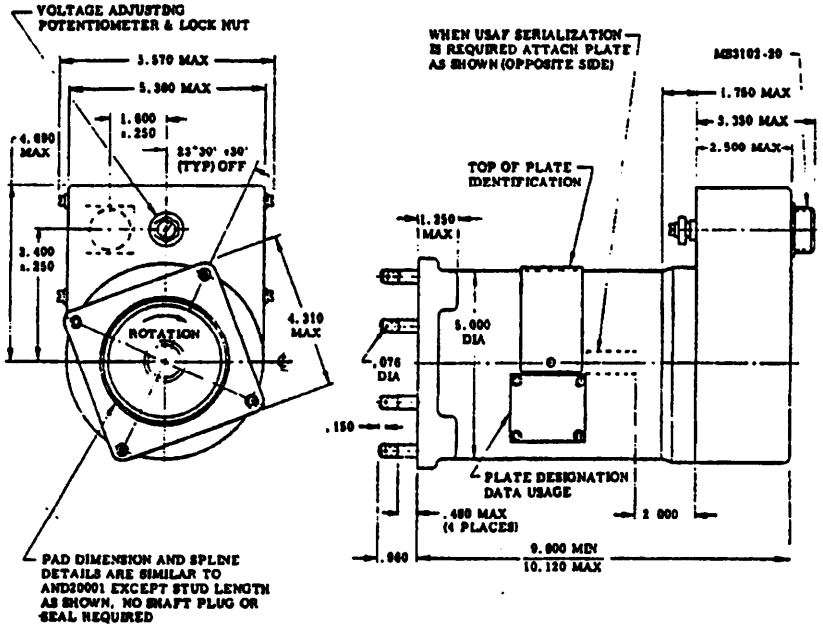
MIL-G-21480/1B(AS)
18 May 1978
Supersedes
MIL-G-21480/1A(AS)
8 November 1968
NOTICE 1
1 July 1976

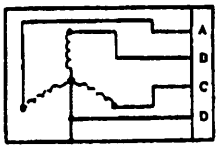
MILITARY SPECIFICATION SHEET

EMERGENCY GENERATOR, ALTERNATING CURRENT, 3.0 KVA, F-4 AIRCRAFT

This specification is approved by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense. The complete requirements for procuring the generator described herein shall consist of this document and the issue in effect of specification MIL-G-21480.



GENERATOR DATA	
RATED VOLTAGE	120/208 V
RATED OUTPUT	3.0 KVA
PHASE	3
SPEED RANGE	12,000 ±5% RPM
OVER SPEED	15,000 RPM
RATED POWER FACTOR	.8 LAGGING TO 1.0
EFFICIENCY	69% MIN AT 3.0 KVA .8 PF
	75% MIN AT 3.0 KVA 1.0 PF
OVERHUNG MOMENT - MAX	95 LB IN
FREQUENCY	380 TO 420 Hz
WEIGHT	20 LBS MAX



PHASE ROTATION
A-B-C

Electric Circuit

PART NUMBER
M21480/1-3

FIGURE 1. GENERATOR

FSC 6115

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REQUIREMENTS:

The requirements of MIL-G-21480 will pertain except for the following additions, deletions, or exceptions.

Delete: All references to supervisory panels.

3.4.2.1 Delete and substitute. Figure 2 of this specification sheet pertains.

DELETE: 2.2, 3.4.6, 3.4.7, 3.4.9.3, 3.4.9.6, 3.4.9.6.1, 3.4.9.6.1.1, 3.4.10.5, 3.4.12, 3.4.12.1, 3.5.2, 3.5.2.1, 3.5.2.2, 3.5.5, 3.5.7, 3.5.7.1, 3.5.7.2, 3.5.7.3, 3.5.7.4, 3.5.7.5, 3.5.7.6, 3.5.7.7, 4.4.1, 4.4.1.1, 4.4.1.1.1, 4.4.1.1.2, 4.4.1.2, 4.4.1.2.1, 4.4.1.3.1, 4.4.1.3.2, 4.4.1.4, 4.4.2.1.2, 4.4.2.3, 4.4.2.3.1, 4.4.4, 4.5.6, 4.5.7, 4.5.8, 4.5.8.1, 4.5.10, 4.5.11, 4.5.15, 4.5.16, 4.5.17, 4.5.18, 4.5.19, 4.5.20.1, 4.5.20.2, 4.5.22.1, 4.5.22.2, 4.5.22.3, 4.5.22.3.1, 4.5.22.3.2, 4.5.22.3.2.1, 4.5.22.3.3, 4.5.23.1.2, 4.5.23.2, 4.5.24.1, 4.5.24.2, 4.5.24.2.1, 4.5.24.2.2, 4.5.24.2.3, 4.5.24.2.4, 4.5.24.2.5, 4.5.24.3, 4.5.24.3.1, 4.5.24.3.2, 4.5.24.3.3, 4.5.24.3.4, 4.5.25.1, 4.5.25.2, 4.5.25.2.1, 4.5.26, 4.5.27, 4.5.28, 4.5.29, 4.5.30, 4.5.31, 4.5.32, 4.5.33, 4.5.33.1, 4.5.33.2.

3.4.8.2 Delete and Add: The generating system shall be capable of delivering a minimum of 100 percent full load at .80 to 1.0 power factor. Generator shall be capable of 167 percent rated load, .5pf for 3 seconds, 107 volts minimum.

3.4.9.4 Delete: "Counterclockwise" and substitute "clockwise".

ADD: 3.4.9.4.1 Rotor acceleration: The generator shall be capable of accelerating from 0 to 12,600 rpm in 0.5 seconds at no load without any detrimental effects.

3.4.9.6.1 Self cooling - Delete and add: The unit shall be self cooled and shall meet specified requirements under the temperature altitude conditions of Figure 2 of MIL-G-21480/1.

3.4.9.7 and 3.4.10.3 Life - Delete and Add: The minimum operating life shall consist of 50 duty cycles plus 1000 hours at no load. A duty cycle shall consist of full load at 0.8 power factor for 15 minutes, with a 67 percent overload for 3 seconds at the beginning of the cycle (after generator has built up to rated voltage).

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3.4.10.1 Delete and Add: Voltage adjust potentiometer required (external).

3.4.10.4 Mounting - Delete and Add: The voltage regulator is mounted integral with the generator.

3.5.1 Add: Voltage Regulation: At any load between 75 percent and 100 percent of full load, the voltage of each phase shall be between the limits of 110 and 120 volts. The no load voltage shall not be greater than 120 volts.

3.5.1.1 Add: Voltage Versus Frequency - The generator phase voltage output shall be within the limits specified in figure 3b over the frequency range indicated with zero electrical load applied.

3.5.3 Phase Displacement and Voltage Unbalance - Delete and Add: The displacement between adjacent phases of the fundamental voltage (excluding harmonics) shall be $120^{o} \pm 5^{o}$. With two phases loaded with 75 percent and the third phase loaded with 100 percent of full load or with one phase loaded with 75 percent and the other two phases loaded with 100 percent of full load the maximum voltage unbalance shall not exceed 4.5. percent.

3.5.4 Change 3 percent to 5 percent.

3.9.1.1 No Load Torque. The torque required to drive the generator at the no load condition shall not exceed 7.8 in-lbs at any rpm from 0 to 12,600.

4.4.2.2 Ambient Temperature - Delete and Add: The ambient temperature shall be $25^{o} \pm 15^{o}$C. No form of cooling other than that integral with the generator shall be used.

4.4.3.2 Warm-up - Delete and Add: There shall be no warm up operating period prior to the tests.

4.4.3.5 Load Conditions - Delete and Add: Load conditions shall consist of full load at 0.8pf. Each phase shall be equally loaded.

4.5.1.1 Failure Criteria - Deterioration or change in performance which could in any manner prevent the equipment from meeting specified functional performanace, require the replacement of components or results in inoperable adjustable components shall provide reason to consider the equipment as having failed to comply with the conditions of the test to which it was subjected.

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4.5.3.1 Overload - Delete and Add: 5 kva at 0.5pf for 3 seconds, 107 volts minimum.

4.5.8.2 Acceptance Test: Change 0.5 percent to 1 percent maximum.

4.5.9 Efficiency: The efficiency of the generator at room ambient temperature and pressure shall not be less than 69 percent at 3 kva, 0.8pf and 75 percent at 3 kva, 1.0pf.

4.5.12 Electromagnetic Interference: Tests shall be conducted in accordance with paragraph 4.5 of MIL-E-81910.

4.5.14 Endurance - Delete and Add: The generator and regulator shall be tested as follows:

(1) Eleven (11) cycles, each cycle consisting of four (4) three hour intervals at altitudes of sea level, 20,000, 40,000, and 60,000 feet respectively. The interval shall be divided into 6 one-half hour periods. The generator shall be loaded to 10 percent during the first and fourth one-half hour periods, 50 percent during second and fifth periods, and 100 percent during the third and sixth periods. All tests are to be run at the maximum temperature shown on Figure 2 of MIL-G-21480/1 with the exception that the tests during one of the intervals at sea level shall be conducted at -40°C. A one-half hour (Minimum) inoperative period shall precede each interval.

(2) Following (1) above the unit shall complete 10 hours of operation on an F-4 aircraft under loaded conditions (cumulative).

(3) Following (2) above test one 3-hour interval as defined in (1) above at sea level and maximum temperature.

4.5.20 Salt Fog - The unit shall be subjected to a salt fog test in accordance with paragraph 4.6 of MIL-E-81910.

4.5.21 Fungus - The generating system shall be subjected to a fungus test in accordance with paragraph 4.3 of MIL-E-81910.

4.5.22 Dust - The unit shall be subjected to a dust test in accordance with paragraph 4.2 of MIL-E-81910.

4.5.23.1.1 Vibration - These items shall be vibrated in accordance with paragraph 4.9 of MIL-E-81910 except that operation during test is not required.

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4.5.24 Shock - The generating system shall be subjected to a shock test in accordance with paragraph 4.7 of MIL-E-81910.

4.5.25 Humidity - The generating system shall be subjected to a humidity test in accordance with paragraph 4.4 of MIL-E-81910.

NOTES:

1. Dimensions in inches. Unless otherwise specified tolerances: decimals ± 0.016.

2. Where only maximum dimensions are shown the part need not have the shape shown but should be contained within the outline shown.

3. Protrusions such as rivet and screw heads shall be contained within the outlines shown.

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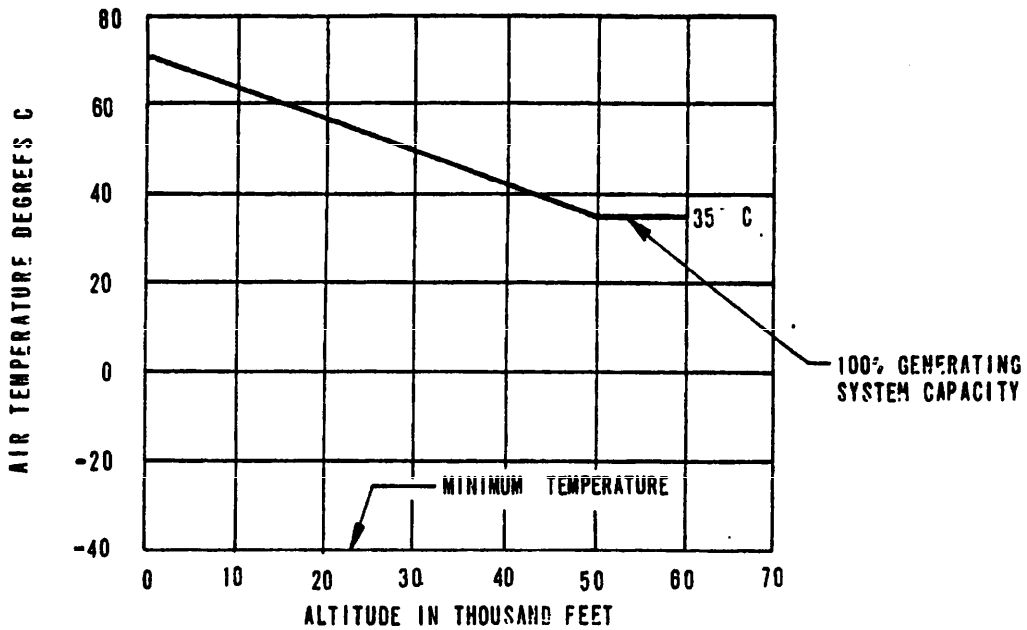


FIGURE 2 TEMPERATURE - ALTITUDE RANGE

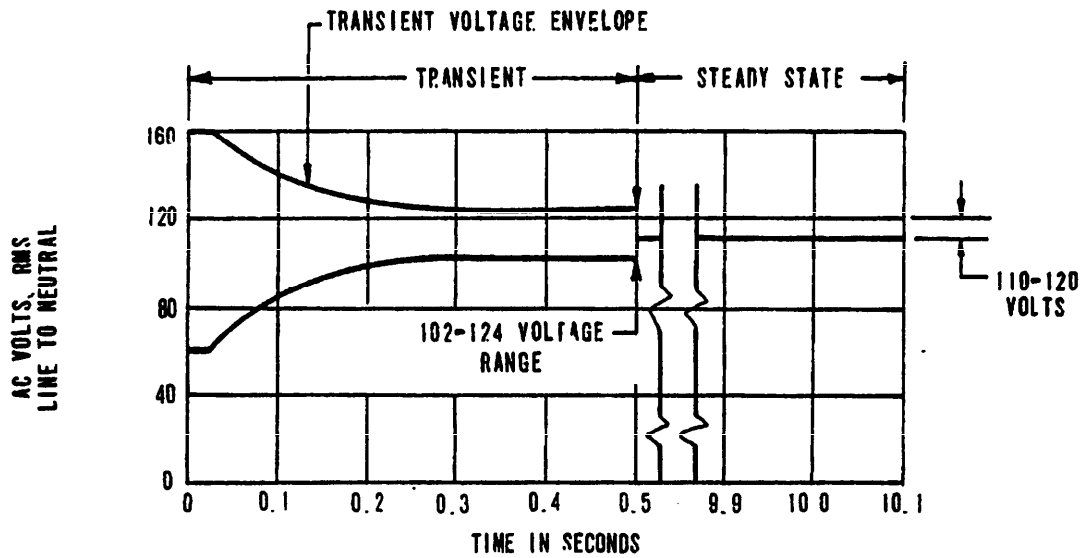


FIGURE 3a TRANSIENT AND STEADY STATE VOLTAGE LIMITS

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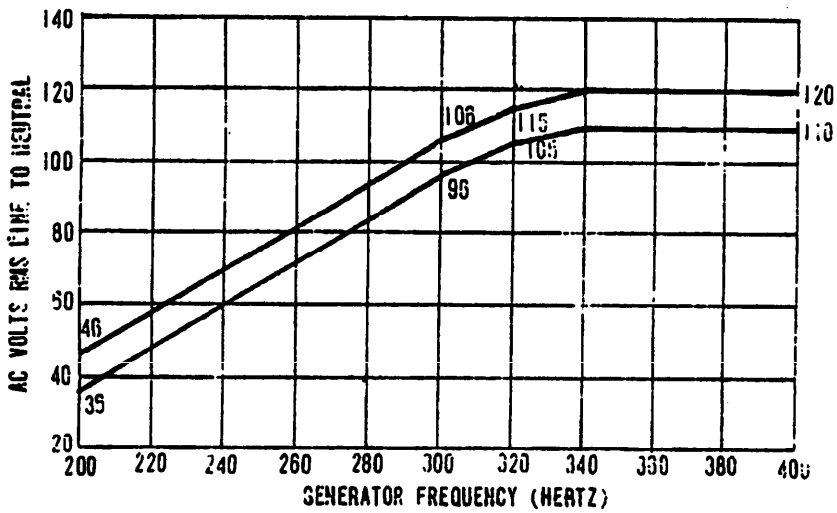


Figure 3b Voltage vs Frequency Limits (No Load)

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